Names: ____________________ ____________________ Table: _____

____________________ ____________________

Discussion Assignment 7

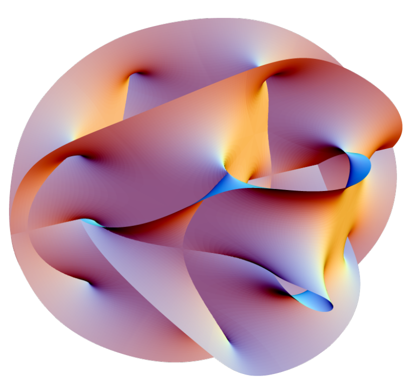
Variable Switcheroo: Electric Boogaloo MATH 2210, Spring 2018

Background

The 1984 hit sequel ”*Breakin’ 2: Electric Boogaloo*” was a break dancing movie featuring Ice-T. In the movie, Ice-T performs a variety of complicated break dancing maneuvers, one of which involves a rapid aerial rotation of legs and hips. For reference, the move resembles what is depicted in the image below.

For decades, many of the world’s top researchers have worked tirelessly, using extremely expensive, highly advanced scientific techniques to determine the smallest smooth shape containing Ice-T’s body during this move. This is a central question in the much-studied mathematical discipline known as ”T-Theory” – which derives its name from the German mathematician Hilbert’s famous *Tanzstellensatz*, or ”theorem of dances.”

In 2017, a team working with the Large Hadron Collider at CERN announced that the problem had been solved – the elusive shape was now known. Interesting and mysteriously, it belongs to a class of objects known as *Calabi-Yau manifolds*, which are conjectured to describe the extra dimensions of spacetime necessary in some formulations of supersymmetric string theory. A visualization of the object in question is given below:



While the precise parameters of this shape cannot be described by elementary functions and require much more advanced theoretical techniques to address, an approximate description is known. Unfortunately, while this approximate description is much easier to work with, it is still quite complicated and is impractical to manipulate without the use of computers. Because of this, we have provided a well-known triple integral which approximates the volume of this enigmatic and mysterious shape, which is as follows:

$$\int_0^3 \int_0^{\sqrt{9-y^2}} \int_{\sqrt{x^2+y^2}}^{\sqrt{18-x^2-y^2}} x^2+y^2+z^2 \;\; dz \;\; dx \;\; dy.$$

In this exercise, we will focus on computing a change of variables for this triple integral. Using the change of variables given by

$$\begin{aligned} x &= r\sin(\phi)\cos(\theta) \\ y &= \rho\sin(\phi)\sin(\theta) \\ z &= \rho\cos(\phi)x^2+y^2+z^2=\rho^2, \end{aligned}$$

compute the resulting form of the integral.

(a) $\displaystyle\int_0^{\pi/4} \int_0^{\pi/2} \int_0^{3\sqrt{2}} \theta\phi^2\cos(\rho) \;\; d\rho \;\; d\phi \;\; d\theta$

(b) $\displaystyle\int_0^{\pi} \int_0^{\sqrt{2}} \int_{\pi/2}^{3\sqrt{2}} \theta\cos(\rho)-\sin(\phi) \;\; d\rho \;\; d\theta \;\; d\phi$

(c) $\displaystyle\int_{\pi/8}^{\pi/4} \int_0^{\pi/2} \int_{-3\sqrt{2}}^{0} (\rho-\phi)^3\tan(\theta) \;\; d\theta \;\; d\rho \;\; d\phi$

(d) $\displaystyle\int_0^{\pi/4} \int_0^{\pi/2} \int_0^{3\sqrt{2}} \rho^4\sin(\phi) \;\; d\rho \;\; d\theta \;\; d\phi$

[0]If you can't tell, the premise of this problem is a late April Fool's joke. While T-theory is a real discipline of mathematics, it deals with the analysis of tree graphs in the context of discrete metric spaces, not with break dancing movies from the 80's. Similarly, while Hilbert was a real mathematician, he never proved a "theorem of dances" – though he did prove a "theorem of zeros" called the *nullstellensatz*, which is the basis of the mathematical discipline known as algebraic geometry. Finally, Calab-Yau manifolds are also real (and, indeed, are important in supersymmetric string theory), but are actually just compact Kähler manifolds with a vanishing first Chern class and a Ricci-flat metric – which in no way has anything to do with break dancing or Ice-T.

Group Work

The group work portion of your grade is worth 3 points. As a group you are allowed to submit three answers. However, incorrect answers will be counted as wrong. For example, if the correct answer is (d) and your group submits $(a),(d),(d)$ as their answers the score will be 2/3 on this portion of the assignment.

Individual Work

Now you are to write a report, one page maximum, that convinces an audience of your peers you have correctly determined the solution to the problem above. This report will need to be saved as a pdf and submitted to your discussion sections WyoCourses page. A scoring rubric and guideline on writing mathematics properly is posted on the WyoCourses page.